

POLICE STATION LOCAL AREA NETWORK PROJECT REPORT

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INTRODUCTION

The criminal activity is increasing now days with high volume, which makes people to feel stressed and unsafe on the streets. As we live in age of technology, this problem must be addressed and improvement, regarding the safety of the people needs to take place and full advantage of the new technology must be taken. There are countries in Europe, that are still using old methods and equipment, which affect the effectivity of the police, to maintain quality and the speed of solving criminal problems. One of those countries is Bulgaria. The police stations in the CAPITAL, are lacking equipment and more particular proper local networks. There, the police officers still use typing machines and lockers for keeping sensitive files. There is no even internet connection. To solve this problem, the project is focused on building suitable and fully configured plan for a local network for police station, that is ready to be installed in the station. This plan, will be created in the form of topology, using software created by Cisco called "Packet Tracer". Also, considering the fact that the local network will be for a police station, just creating it is not going to be enough, the security must be taken seriously, which mean that there is going to be a security policy in place, to enforce important rules. Any bridge, any information leaking, and vulnerabilities can be fatal for the police.

Before creating the network, all components must be pointed for better understanding about, how network works. Therefore, the first pages are about the enterprise network itself, network devices, network media etc.

LITERATURE REVIEW

This literature review seeks to offer a summary of the current academic research and works in the area of networking. The main goal is to discover important networking trends, technologies, and difficulties that might guide a networking project. Network protocols, network security and wireless networking are just a few of the many subjects covered in the examination. The planning and execution of the networking project will be built on the conclusions of this literature research.

1. Introduction

- Definition and importance of networking
- Overview of the networking project

2. Network Protocols

- TCP/IP protocol suite
- Ethernet
- Routing protocols (e.g., OSPF, BGP, RIP)

3. Network Security

- Firewalls and network access control

4. Wireless Networking

- Wireless LAN (Wi-Fi) standards (e.g., 802.11ac, 802.11ax)
- Internet of Things (IoT) networking

5. Network Virtualization

- Virtual LANs (VLANs)
- Network orchestration and automation

6. Network Management

- Network monitoring and troubleshooting
- Network performance optimization
- Network configuration management

7. Challenges and Future Directions

- Scalability and performance
- Network resilience and reliability
- Energy-efficient networking

10. Conclusion

- Summary of key findings
- Implications for the networking project

This survey of the literature in the area of networking offers a thorough overview of the pertinent literature. It covers a range of networking topics, such as wireless networking, protocols and security. The assessment analyses problems in the networking industry and spotlights new developments. The networking project's development and execution will be informed by the results of this literature assessment, allowing it to take advantage of the most recent developments and industry best practises.

ENTERPRISE NETWORK

The interconnected system of computers, gadgets, and communication infrastructure that makes up a corporation is referred to as an enterprise network. It is made to make it easier for staff members, departments, and locations throughout a corporation to communicate, share data, and access resources with ease. Enterprise networks are essential for providing information flow that is both efficient and secure, fostering collaboration, and supporting business processes.

The key components of an enterprise network typically include:

- 1. Local Area Network (LAN).** The LAN serves as the building block of an enterprise network and links devices that are located within a certain geographic region, such as a campus or office building. It makes it possible for computers, printers, servers, and other networked devices to communicate quickly and effectively.
- 2. Wide Area Network (WAN).** The WAN increases network connection outside of a single site. It enables businesses to link several locations across a wider geographic region, including branch offices, data centres, and remote employees. Leased lines, MPLS (Multiprotocol Label Switching), and VPN (Virtual Private Network) are WAN technologies that provide fast and secure data flow between various sites.
- 3. Network infrastructure.** The hardware and software elements that provide network connectivity and data transfer are referred to as network infrastructure. It consists of network cables, wireless access points, switches, routers, and firewalls. The configuration, monitoring, and optimisation of the network infrastructure are controlled by software and protocols for network management.
- 4. Internet Connectivity.** Internet connectivity is necessary for the majority of business networks in order to access external resources, cloud services, and international communication channels. The connection is provided by Internet Service Providers (ISPs), and enterprises frequently use security tools like firewalls and proxy servers to shield their networks from outside threats.

5. Security and Access Control. Because sensitive data is handled by enterprise networks in large quantities, security is a key concern. Firewalls, intrusion detection/prevention systems, and encryption protocols are examples of network security techniques that protect against malware assaults, illegal access, and data breaches. Only persons with the proper authorization can access certain resources and information thanks to access control technologies like user authentication and role-based permissions.

6. Network Services. The effectiveness and functionality of a corporate network are improved by various network services. These services include the Network Time Protocol (NTP) for time synchronisation between network devices, the Dynamic Host Configuration Protocol (DHCP) for automatic IP address allocation, and the Domain Name System (DNS) for converting domain names into IP addresses.

7. Network Management. The company network must be monitored, configured, and troubleshooted using effective network management. Administrators may manage network resources, monitor performance indicators, find and fix problems, and put security rules into place with the help of network management tools and protocols.

For businesses to maintain their competitive edge, simplify operations, and allow seamless communication and collaboration across their entire infrastructure, they need an enterprise network that is reliable and well-designed. [1] [2]

LOCAL AREA NETWORK

A local area network (LAN) is a type of computer network that mostly exists within a single building or campus and covers a relatively small geographic region. It enables the sharing of resources like files, printers, and internet connections between computers and other devices. LANs are often used in homes, workplaces, educational institutions, and other small-scale settings.

LANs are intended to offer dependable, high-speed communication between nearby devices. Though wireless LANs (WLANs) have gained popularity as a result of the development of Wi-Fi technology, they are often constructed utilising wired connections, such as Ethernet cables or fibre optic cables.

Computers, servers, switches, routers, and network cables are the essential elements of a LAN. Through network ports on switches or routers, computers and other devices, such as printers or network storage devices, are linked to the LAN. While routers allow communication between the LAN and external networks, such as the internet, switches act as hubs for connecting devices inside the LAN.

The ability to share resources is one of a LAN's main benefits. For instance, a LAN enables effective cooperation and data sharing since numerous computers may access a common printer or a central file server. A safe environment for data transit within a closed network is another benefit of LANs, which lowers the possibility of unauthorised access from outside sources.

For improved speed and security, LANs can be further separated into subnets or virtual LANs (VLANs). While VLANs form logical groupings inside a LAN, subnets divide a LAN into smaller networks, providing greater granular control over network traffic and security.

LANs are now an essential component of contemporary computing, enabling collaboration, resource sharing, and communication in intimate settings. They have also opened the way for the creation of bigger networks like the internet and WANs, which link LANs together across great distances to enable global connection. [3]

WIDE AREA NETWORK

A computer network that connects several local area networks (LANs), as well as other devices, is known as a wide area network (WAN). It offers the infrastructure necessary for people and organisations to communicate and exchange resources over great distances.

WANs link devices and networks that are physically apart by enabling the transfer of data, audio, and video across a large scale. To provide dependable and secure communication across diverse locations, including offices, branches, and data centres, they make use of a variety of technologies and protocols.

Connecting LANs in various places, frequently across public or private telecommunications networks, is one of the main functions of WANs. Through the creation of a unified network environment, firms are able to centralise management, efficiently share resources, and expedite communication.

Leased lines, circuit-switched networks, packet-switched networks, and virtual private networks (VPNs) are only a few of the communication technologies used by WANs. With the use of these technologies, data can be transmitted quickly and securely in a variety of formats, including text, audio, and video.

WANs are becoming even more crucial in the connected world of today with the introduction of cloud computing, the rise of remote access, and distributed workforces. They make it easier to integrate cloud services, distant users, and branch offices into one integrated network architecture.

By enabling organisations to duplicate data across many locations, WANs also play a significant role in data replication and disaster recovery, assuring data availability and business continuity in the event of a network failure or natural catastrophe.

Wide area networks have fundamentally changed how people and organisations communicate, work together, and access resources over great distances. As technology develops, they continue to change to meet the rising needs of global connection and make a variety of applications possible that promote development, creativity, and productivity. [4]

NETWORK INFRASTRUCTURE

The underlying framework of a network that facilitates data transmission and communication between devices, systems, and users is referred to as network infrastructure. It is made up of a sophisticated ecosystem of hardware, software, protocols, and technologies that enable smooth data transfer and communication through local, wide-area, and international networks.

At its heart, network infrastructure consists of a number of parts that cooperate to provide effective and dependable communication. These elements consist of networking equipment such as routers, switches, servers, cables, wireless access points, firewalls, load balancers, and more. Each component is essential to permitting information flow and preserving the network's integrity and security.

By routing data packets across various networks to their intended destinations, routers serve as traffic directors. On the other hand, switches allow numerous devices to join to a local area network (LAN) and interact with one another. Users and devices may easily access data storage, applications, and network resources thanks to servers, which act as the centre for these activities.

Data is transported through cables as the physical medium, both fibre optic and wired. With fibre optic cables enabling increased capacity and longer transmission lengths, these cables guarantee dependable and fast connectivity. Mobility and flexibility are made possible by wireless access points, which extend network connectivity to wireless devices.

Firewalls are used to protect network infrastructure from unauthorised access and possible threats. In order to prevent malicious actions, firewalls apply security controls to both incoming and outgoing network traffic. By distributing network traffic among several servers, load balancers improve performance and guard against overloading.

Standards and protocols lay the groundwork for network communication. While Transmission Control Protocol (TCP) assures dependable data transmission, the Internet Protocol (IP) facilitates the routing of data packets over the internet. In some areas of network operation, other protocols like Ethernet, Wi-Fi, and DNS (Domain Name System) play crucial roles.

Wide area networks (WANs) and global networks, which link geographically scattered places, are examples of network infrastructure that are not restricted to a single location.

In conclusion, network infrastructure is the foundation of contemporary connection, allowing for smooth data and communication transfer across diverse devices, systems, and places. Its ongoing development and acculturation to new technologies are essential in order to satisfy the growing needs of our networked environment. [5]

INTERNET CONNECTIVITY

With its seamless integration into our social, professional, and personal lives, internet connectivity has grown to be an essential component of our everyday existence. The way we interact, study, work, and enjoy ourselves has been completely transformed by our capacity to access the massive network of information and services that the internet offers.

The internet was once a luxury enjoyed by a select few, but because of technical breakthroughs and greater attempts to improve global connection, it is now progressively becoming more available to individuals all over the world. Today, billions of people use the internet for a wide range of activities, from straightforward site surfing and social media communication to more involved activities like cloud computing, online gaming, and remote collaboration.

The complex global network architecture is the cornerstone of internet connection. It includes a large network of satellites, cellphone towers, fiber-optic cables, and other communication infrastructure. The backbone of the internet is made up of these physical elements as well as numerous routers, switches, and servers, allowing data to move across great distances at remarkable speeds.

In many industrialised nations, broadband internet is now considered standard, giving customers high-speed, always-on access. This has sparked a digital revolution, making it possible for information to be exchanged quickly and for innumerable new applications and services to develop. The internet has revolutionised a number of sectors and enabled people to connect, produce, and interact in ways that were previously unthinkable. From video streaming platforms and e-commerce websites to teleconferencing tools and cloud-based storage solutions.

It's crucial to recognise that internet access is still not widespread and that there are still digital inequalities, especially in underserved rural communities and developing regions. Through programmes like satellite-based internet services, neighbourhood networks, and public-private partnerships, efforts are being made to close this gap. The objective is to guarantee that everyone has

equal access to the wealth of information and possibilities that the internet has to offer, irrespective of their location or socioeconomic level.

Even more opportunities and difficulties lie in store for internet connectivity in the future. With higher speeds, reduced latency, and more dependable connections, emerging technologies like 5G networks, satellite constellations, and mesh networks have the potential to substantially improve connectivity. The emergence of smart cities and the proliferation of Internet of Things (IoT) devices will also put unprecedented demands on internet infrastructure, needing ongoing improvements in network capacity and security.

The globe has become a global village thanks to internet connectivity, which has eliminated geographic borders and linked individuals from different backgrounds. It has increased economic growth, empowered individuals, and stimulated innovation in a variety of industries. In order to build a more inclusive and connected society for everyone as we go forward into a digitally driven future, it is essential that we continue to prioritise and invest in increasing and strengthening internet access. [6]

SECURITY AND ACCESS CONTROL

Security and access control are now top priorities for everyone in the modern world, including people, businesses, and governments. Access control systems and strong security measures have been developed as a result of the necessity to safeguard private data, priceless assets, and people's safety.

The term "security" refers to a broad variety of procedures and tools used to prevent unauthorised entry, data breaches, theft, vandalism, and other possible dangers. It entails setting up layers of defence that serve as barriers to stop unauthorised people or groups from accessing secure locations, systems, or data. Physical barriers like fences, gates, locks, and security cameras are only a few examples of these security measures. Digital protections like firewalls, encryption, and intrusion detection systems are also options.

On the other hand, access control specialises in controlling and managing the entrance and exit points to a certain region or system. It entails deciding who is permitted access to what data, places, or resources and under what restrictions. These limits are enforced by access control systems using a variety of authentication approaches, including passwords, PINs, smart cards, biometric scans (such as fingerprint or face recognition) or a combination of these methods. Organisations may maintain a high degree of security by establishing access control methods that make sure that only authorised people are given access to sensitive locations or secret data.

Access control and security are significant across many different industries. It aids in the physical protection of vital infrastructure, governmental buildings, commercial buildings, and private assets against unauthorised access, theft, or destruction. Office buildings, airports, hospitals, data centres, and research labs all often utilise access control systems to manage and monitor pedestrian traffic and safeguard priceless assets.

Access control and security are essential in the digital world to protect sensitive data that is held in databases, computer networks, and online platforms. To safeguard their digital assets from cyberthreats like hacking, malware, and data breaches, organisations rely on sophisticated security

methods like encryption algorithms, multi-factor authentication, and frequent security audits.

Effective security and access control measures need a multifaceted strategy that incorporates human, technical, and physical components. It entails creating strong security policies, putting in place suitable controls, instructing staff in best practises, and keeping up with the most recent security trends and dangers. Organisations must also routinely analyse and evaluate their security defences to spot weaknesses and take preventative measures to reduce risks.

In conclusion, security and access control are essential for protecting tangible and digital assets, securing private data, and assuring user safety.

Organisations can maintain a safe environment, cultivate trust with their stakeholders, and reduce possible dangers in a world that is becoming more linked by putting robust security practises and access control measures in place. [7]

NETWORK SERVICES

The numerous technologies, protocols, and applications that provide data transport and communication through computer networks are referred to as network services. These services are essential for linking devices, streamlining data transfer, and enabling information to move freely between various network settings. In this section, we'll examine several crucial elements of network services and their significance for contemporary connection.

The capacity of network services to create and sustain connections is one of its key features. The cornerstone of internet communication is built on protocols like TCP/IP (Transmission Control Protocol/Internet Protocol), which enable devices to connect, exchange data, and guarantee accurate delivery. These protocols control functions including addressing, routing, and error detection, ensuring that data is sent between network nodes reliably and effectively.

DNS (Domain Name System), which converts domain names into their matching IP addresses, is yet another crucial network function. Instead of having to remember lengthy numerical IP addresses, DNS enables users to connect to websites, send emails, and connect to a variety of online services using their familiar domain names. The decentralised and hierarchical name system offered by DNS makes it possible to navigate the internet with ease.

Various security measures are also included in network services in order to safeguard network infrastructure and data. For instance, firewalls serve as gatekeepers by keeping an eye on and filtering incoming and outgoing network traffic in accordance with set criteria. They aid in guarding sensitive information, detecting possible risks, and preventing unauthorised access.

Services for network administration and optimisation are also included. Administrators may use network monitoring tools to keep an eye on network performance, spot bottlenecks, and conduct proactive troubleshooting. Quality of Service (QoS) techniques give network traffic a higher priority in order to better utilise available bandwidth and maintain seamless functioning of mission-critical applications.

In general, network services constitute the foundation of contemporary connection, providing smooth data movement, communication, and access to

online resources. These services are essential to maintaining the efficiency of our linked world because they establish network connections, provide security, and improve performance. [8] [9]

NETWORK MANAGEMENT

The process of monitoring and regulating the use, maintenance, and administration of computer networks is referred to as network management. Effective network administration is essential for guaranteeing seamless and dependable communication in today's linked world, as networks play a significant role in linking people, devices, and services.

Network administration includes a wide range of duties and responsibilities, such as capacity planning, performance optimisation, security management, and network monitoring. To maintain network operation, improve performance, and reduce possible problems, a combination of hardware, software, and human skills is used.

Network monitoring is one of the core tenets of network management. In order to do this, the network infrastructure must be continually monitored, traffic patterns must be examined, and any irregularities or performance bottlenecks must be found. Administrators can quickly discover and resolve issues thanks to the real-time insights monitoring tools give them into the state of their networks.

Optimisation of performance is yet another crucial aspect of network administration. It entails fine-tuning network resources to maintain optimal performance and prioritise crucial applications or services, such as bandwidth allocation and quality of service (QoS) setups. Organisations can increase productivity, lower latency, and offer a smooth user experience by optimising network performance.

Given the evolving threat landscape and the need of protecting sensitive data, security management is a crucial component of network administration. It entails putting in place strong security measures to safeguard the network against unauthorised access, data breaches, and other cyber threats. Examples include firewalls, intrusion detection systems, and encryption methods. Patch management, vulnerability assessments, and security audits on a regular basis are necessary to ensure network security.

Network management inevitably involves troubleshooting. Networks are intricate systems, and problems can occur for a number of reasons, such as

hardware malfunctions, incorrect settings, or software incompatibilities. To reduce downtime and restore network operation, network administrators are in charge of quickly detecting and fixing these problems.

Planning for capacity is essential for predicting future network needs and guaranteeing scalability. To predict the network's future requirements, network administrators examine historical data, growth predictions, and cutting-edge technology. Organisations may prevent traffic jams, optimise resource allocation, and make wise choices about network improvements or expansions by precisely estimating capacity demands.

With the introduction of new technologies like cloud computing, virtualization, and the Internet of Things (IoT), network administration has grown increasingly sophisticated recently. Additional difficulties have been created by these developments, including managing hybrid settings, guaranteeing secure remote access, and controlling the explosive proliferation of connected devices. Professionals in network administration must adjust to these developments and learn the essential skills to efficiently manage contemporary network infrastructures.

In general, network management is essential for preserving the reliability, efficiency, and security of computer networks. Organisations may achieve dependable connection, effective resource utilisation, and a solid basis for their digital operations with the right management practises in place. [10]

NETWORK DEVICES

Modern communication systems rely heavily on network devices to enable efficient data flow between computers, servers, and other connected devices. These components provide the transfer of information over multiple physical and virtual channels, acting as the foundation of computer networks. Every network component, from routers and switches to firewalls and modems, has a distinct role and enhances the overall effectiveness and security of the network.

1. Routers. Routers are crucial network components that link different networks and route data packets between them. They choose the most effective route for data transmission, making sure that information effectively reaches its target location. Network address translation (NAT) services are another function that routers offer, enabling several devices to share a single internet connection.

2. Switches. Switches are components that allow several devices to connect to a local area network (LAN). They function at the network's data link layer, employing MAC addresses to direct data packets to the desired location. By establishing dedicated paths for data transmission and removing network congestion and collisions, switches improve network performance.

3. Firewalls. A network's first line of defence against intrusive access and malicious activity is its firewall. In order to prevent unauthorised access, virus infections, and data breaches, these devices enforce security policies by monitoring incoming and outgoing network traffic. A secure barrier between internal networks and the outside world is provided by firewalls, which can be implemented as hardware or software.

4. Modems. To facilitate connection between computers and the internet, modems modulate and demodulate analogue and digital data. They either turn digital signals into broadband signals used in cable or DSL connections or they convert digital signals into analogue signals for transmission over conventional telephone lines (as in dial-up connections). Devices may access the internet through wired or wireless connections thanks to the common integration of modems into routers.

5. Wireless Access Point. WAPs, sometimes referred to as wireless routers, allow for wireless communication within a network. They serve as central nodes that provide network connections and internet service access for gadgets with Wi-Fi capabilities. WAPs employ radio signals to send and receive data, giving computers, cellphones, and other wireless devices easy access to the internet.

6. Network Switches. Similar to LAN switches, network switches bring many networks together at a higher level of operation. To find the fastest pathways for data transmission, they employ sophisticated algorithms, providing the best performance and minimising delay. When connecting numerous LANs, network switches are frequently employed in large-scale business networks.

The establishment and upkeep of contemporary computer networks depend on a variety of network devices, of which these are only a few examples. Each device has a specific role and adds to the network infrastructure's overall functionality, security, and performance. Network devices will adapt and develop as technology develops more, becoming more crucial in our networked society. [11] [12]

NETWORK CABLES

Modern communication relies on network cables, which act as the actual physical conduits for the smooth transfer of data between devices. These cables connect computers, servers, routers, and other network equipment to enable the transfer of information over local area networks (LANs), wide area networks (WANs), and the worldwide internet. They constitute the foundation of computer networks.

The Ethernet cable, usually referred to as the twisted pair cable, is one of the most widely used network cables. It is made up of several pairs of insulated copper wires that have been twisted together to lessen electromagnetic interference. There are several different types of Ethernet cables, including Cat5e, Cat6, and Cat7, each with unique requirements for bandwidth and data transmission rates. RJ-45 connectors are frequently used on these lines to create a dependable connection.

Coaxial cable is an additional variety of network cable. It has a metallic shield, an outer protective sheath, and a centre conductor wire enclosed in insulation. The enormous bandwidth characteristics of coaxial cables make them perfect for carrying cable television signals, internet access, and other applications that call for high data throughput.

The most advanced type of network cable is fibre optic cable. They use light pulses to transport data via glass or plastic fibres. These cables are essential for high-bandwidth applications and long-distance communications because of their excellent speed, dependability, and security. Since electromagnetic interference has no effect on fibre optic connections, data may be sent across great distances without suffering signal deterioration.

Network cables are essential for maintaining constant connectivity and efficient data transfer. In addition to homes, workplaces, data centres, and communications infrastructures, they are employed in many more contexts. The contemporary digital environment on which we rely on would not be feasible without dependable network cables.

Wireless connections are more and more common as technology develops. However, network cables are still necessary for creating reliable and secure

connections, especially in settings where dependability and fast data rates are critical. The basic network cable continues to be the foundation of our networked world, enabling the smooth movement of information that powers our digital lives, whether it's a tiny home network or a sophisticated business setup. [13] [14]

CISCO PACKET TRACER

A robust network simulation tool that offers a virtual environment for studying and applying networking topics is Cisco Packet Tracer. It was created by Cisco Systems to help scholars, teachers, and network specialists comprehend and visualise different network scenarios.

Users may build, manage, and debug virtual network topologies using Cisco Packet Tracer by utilising a variety of Cisco hardware, including routers, switches, firewalls, and wireless access points. Users may connect devices together, change their characteristics, and drag and drop devices into the workspace using the tool's user-friendly interface.

Cisco Packet Tracer's capacity to replicate network behaviour is one of its key advantages. Users may see how packets move over the network and test their setups to see the effects of their modifications. It is a great tool for both novice and expert users since it offers real-time feedback and visual representations of network activities.

Networking technologies supported by Cisco Packet Tracer include Ethernet, Wi-Fi, VLANs, WAN connectivity, and security protocols. Users may practise implementing networking fundamentals like routing protocols (like OSPF, EIGRP) and network services (like DHCP, DNS) by simulating actual-world settings.

The application also provides a substantial collection of tutorials, practise exercises, and learning resources that may be used for networking classes and independent study. These tools support users' skill and knowledge growth in a systematic and dynamic way, promoting a better comprehension of networking concepts.

Additionally, Cisco Packet Tracer offers teamwork, enabling several people to collaborate on a network project at once. This function encourages cooperation and makes it easier for users to share information and ideas.

In conclusion, Cisco Packet Tracer is a useful tool for network training and teaching. Users may create, configure, and visualise network topologies with it. They can also simulate network behaviour and practise using different

networking technologies. Cisco Packet Tracer may be a great ally in your quest to grasp networking, whether you're a student, professor, or network professional. [15]

NETWORK DEVICES CONFIGURATION

Setting up and adjusting a number of features and characteristics on a network device entail coordinating the device with the needs of the network. This procedure often entails gaining access to the device's management interface, which might be either a web-based GUI or a command-line GUI. The following factors should be taken into account while setting network devices:

- 1. IP Addressing.** The setup of network devices begins with assigning IP addresses. For communication to be possible, every device on the network needs a different IP address. The device's IP address, subnet mask, default gateway, and, in certain circumstances, DNS server settings must all be configured.
- 2. Interfaces and Ports.** Network devices may connect to other devices using a variety of interfaces and ports. These interfaces must be configured by indicating factors including speed, duplex mode, VLAN membership, and access control lists (ACLs) to manage traffic.
- 3. Routing.** Routers are in charge of transmitting data between various networks. The optimum path for data transmission may be found by routers by configuring routing protocols like RIP (Routing Information Protocol) or BGP (Border Gateway Protocol).
- 4. Switching.** By linking devices on the same subnet, switches make it easier for devices on a local network to communicate with one another. VLAN (Virtual Local Area Network) configuration covers creating VLANs, allocating ports to VLANs, and configuring trunk ports for inter-VLAN communication.
- 5. Security.** In today's linked world, network security is crucial. Virtual private networks (VPNs), firewall rules, and access control lists (ACLs) may all be configured to assist protect a network against intruders, threats, and data breaches.
- 6. Wireless Access Points.** In order to set up wireless access points (APs) and offer safe and dependable wireless connection, it is necessary to configure SSIDs (Service Set Identifiers), security protocols (such as WPA2, WPA3), channel selection, transmit power, and other wireless-specific parameters.

A thorough grasp of networking concepts, protocols, and device-specific settings is necessary for network device configuration. In order to guarantee that devices function properly and securely, it's crucial to adhere to best practises, record modifications, and often update firmware.

Finally, it should be noted that there are several duties involved in network device configuration, including IP addressing, interface and port configuration, routing, switching, security, QoS, wireless setup, and monitoring. A stable, secure, and effective network infrastructure is produced when network devices are configured properly to guarantee they perform as intended. [16]

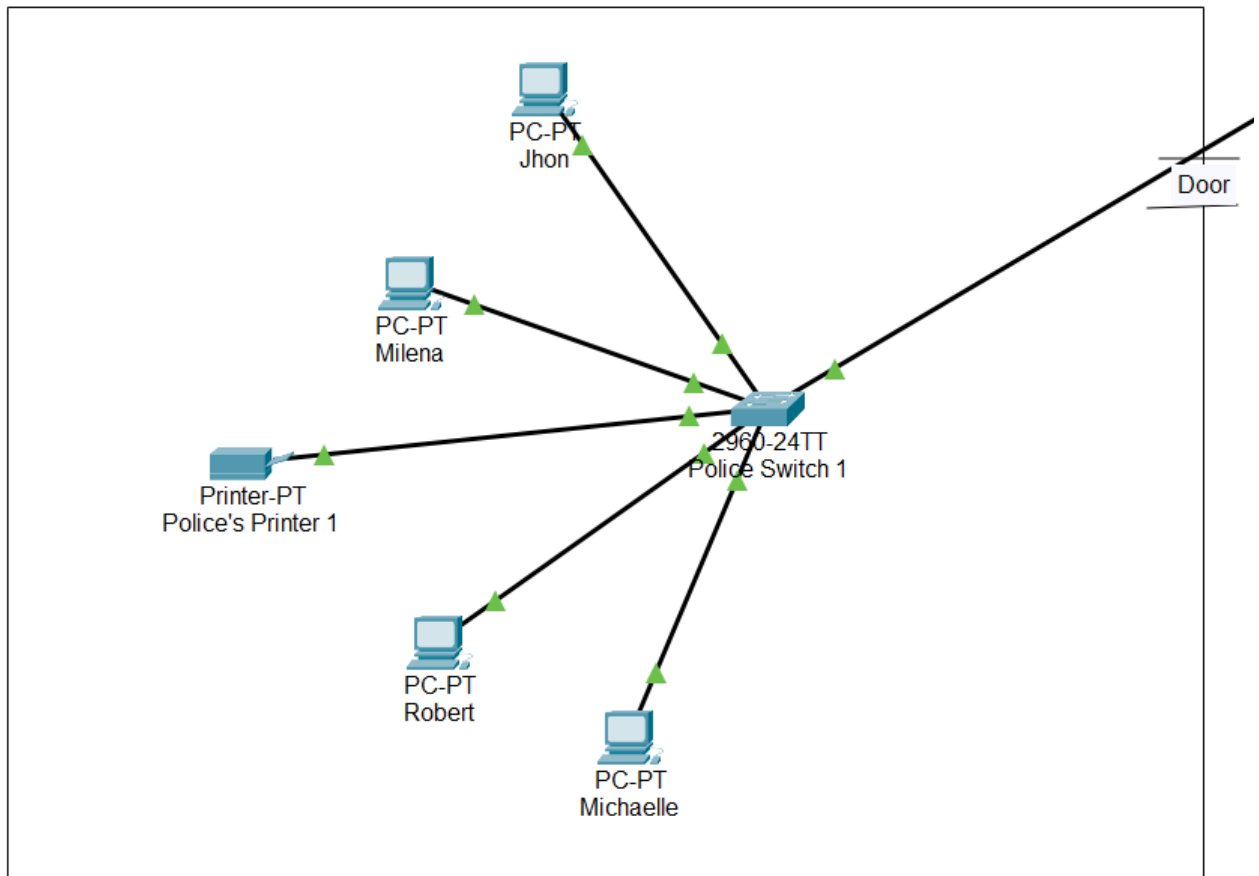
METHODOLOGY

After all that has been said about the enterprise network, it is clear that a local network is just a part of it, in fact, one big network is formed from several small networks and several big networks is what we call enterprise network. The internet is literally hundred thousand networks connected with each other. This project is one local network that it is a small part of the internet, for now logically and hopefully physically in the future. The aim of this network is to help improve the productivity and the quality of the police stations in Bulgaria. To aim this project, an interview took place before the start of this network plan, with the police inspector Iliya Atanasov from the fifth local station in the capital of Bulgaria Sofia. During the interview, he explained about the daily duties in the station and about the poor equipment that they have to work with. **This project network topology is created based on what he explained about the station needs and functionality, it is not just randomly made topology.** In the interview, he speaks about the lack of equipment like, computers, laptops, Wi-Fi, data storage, communication, server and so on. He said that in the station, they still use a typing machine and store data in a locker. There is no Wi-Fi and proper communication between the departments. Based on this, the aim of this project is to build a proper local network for this police station and bring it to the 21st century. This is going to increase an important field in the country, the fight against crime. The officers are going to be faster and more productive, and the information is going to be much more secure than before.

The building where the local network will be is divided into seven rooms, or seven departments. The departments are **two officer's rooms**, where the regular officers can do their duties, **one interrogation room**, where the suspect will be kept for investigations, **one criminal room**, for more advanced research, **one lab**, for fingerprint etc, **administrator room** and **server room**.

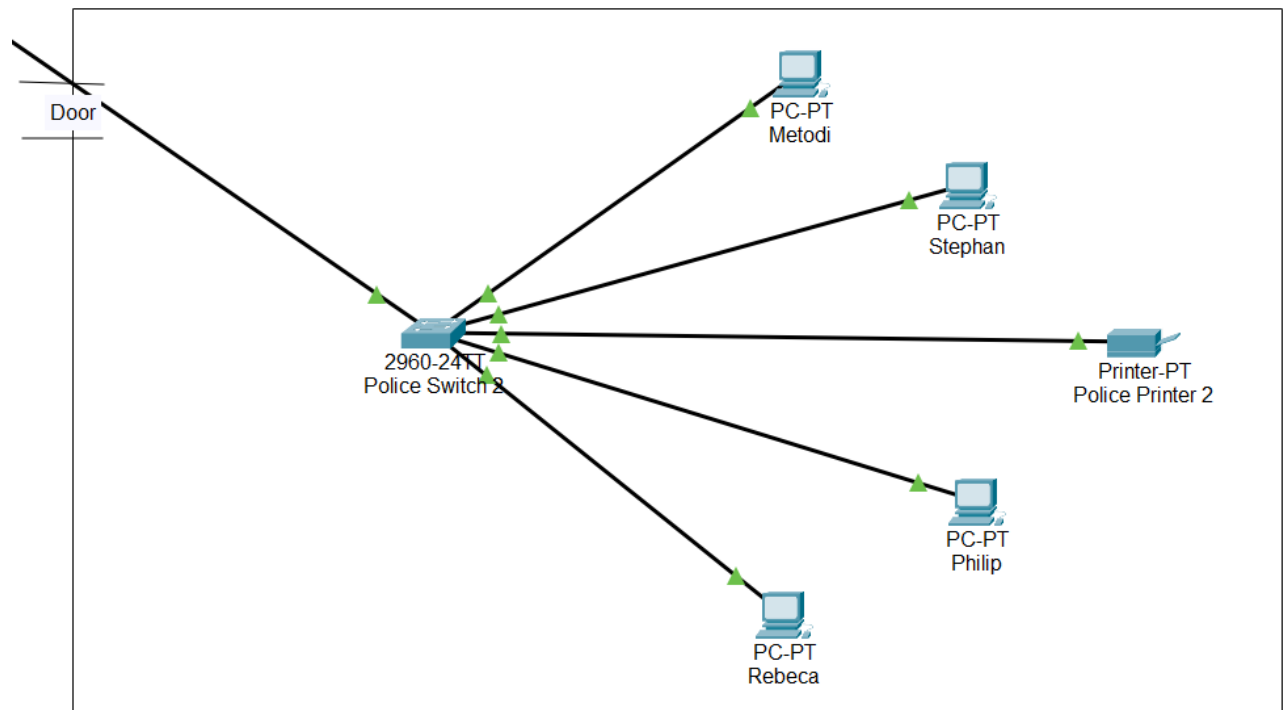
OFFICER'S ROOMS

The rooms are identical one another, the first one looks like this:

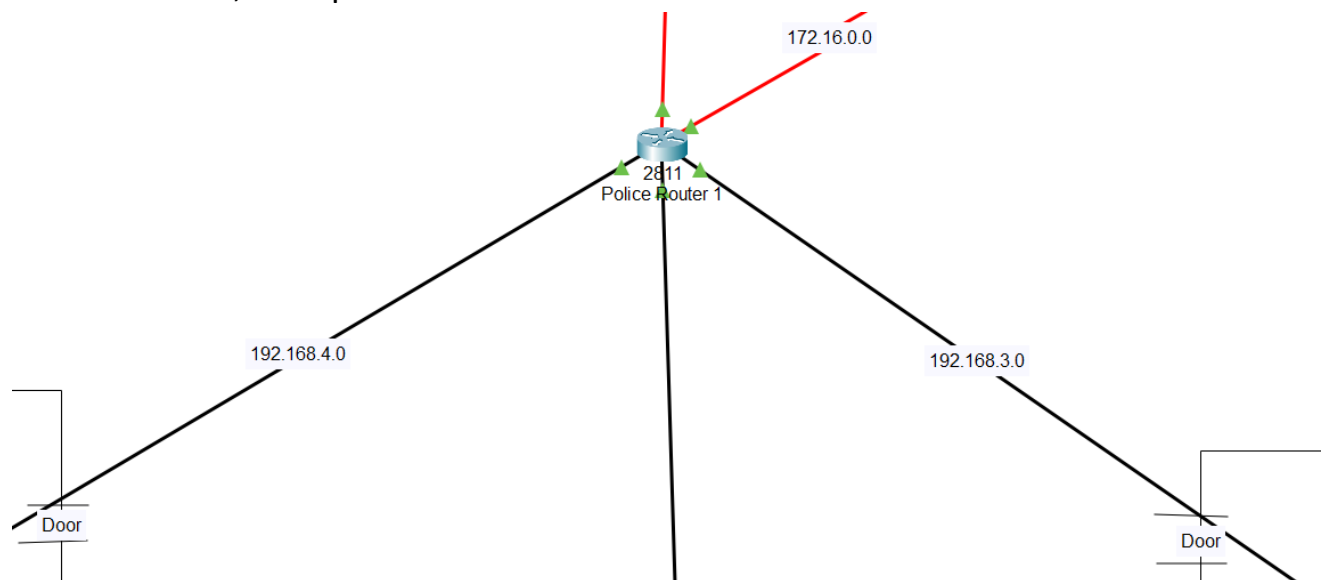


All devices are connected to each other via switch, this switch is able to connect up to 24 devices and it is suitable in case if more devices need to be added. There is also a printer, connected via cable and configured with an IP address. The computers and the printer are on 192.168.4.0 network. All devices are configured with a static IP address.

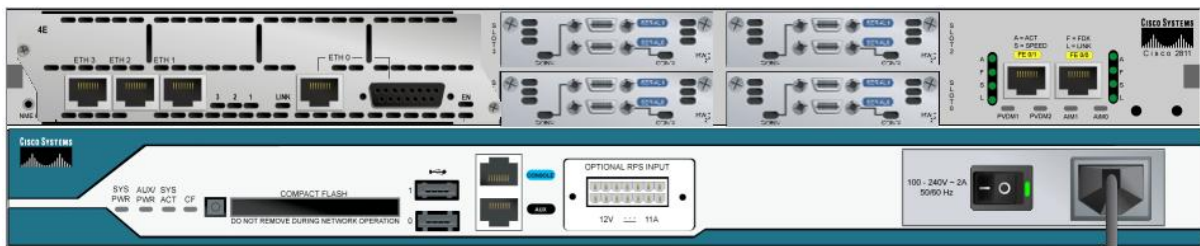
The second, room is same as the first one apart from the network. This network is 192.168.3.0.



Both networks, or departments are connected with each other via router.



This is Cisco 2811 router.



This router is suitable for connecting multiple networks, due to the possibility of installing more interfaces on it. This is why was chosen. Each interface is configured with static suitable IP address, the router runs with RIP (Routing Information Protocol) routing protocol. This routing protocol is suitable for small networks and easy for managing.

Configuring a Cisco router, such as the Cisco 2811, involves a series of steps to set up its basic settings, interfaces, and routing capabilities.

1. Access the Router.

Connect to the router using a console cable and terminal emulation software (e.g., PuTTY, HyperTerminal). Access the router's command-line interface (CLI).

2. Enter Privileged EXEC Mode.

After logging in, enter privileged EXEC mode:

“enable”.

3. Enter Global Configuration Mode.

Enter global configuration mode to make configuration changes:

“configure terminal”.

4. Hostname.

Set the hostname for the router:

“hostname Police Router 1”. Changes the router’s name.

5. Interface Configuration.

Configure router interfaces. For example, configuring IP addresses on interfaces:

“interface GigabitEthernet0/0”. Enter the particular interface.

“ip address 192.168.4.1 255.255.255.0”. Assign an IP address to this interface.

“no shut” It switches the interface from off to on.

6. Assigning Interfaces to VLANs (For further need):

“interface GigabitEthernet0/0/0.10”. Defines the VLAN 10 on this interface.

“encapsulation dot1Q 10”. Enables 802.1Q protocol for VLAN 10.

“ip address 192.168.10.1 255.255.255.0”. Assign an IP address for the interface and default gateway for the devices connected to it.

7. Assigning routing protocol:

“router rip”. Enters the RIP protocol configuration.

“network 192.168.4.0”. Advertising a network connected to this router.

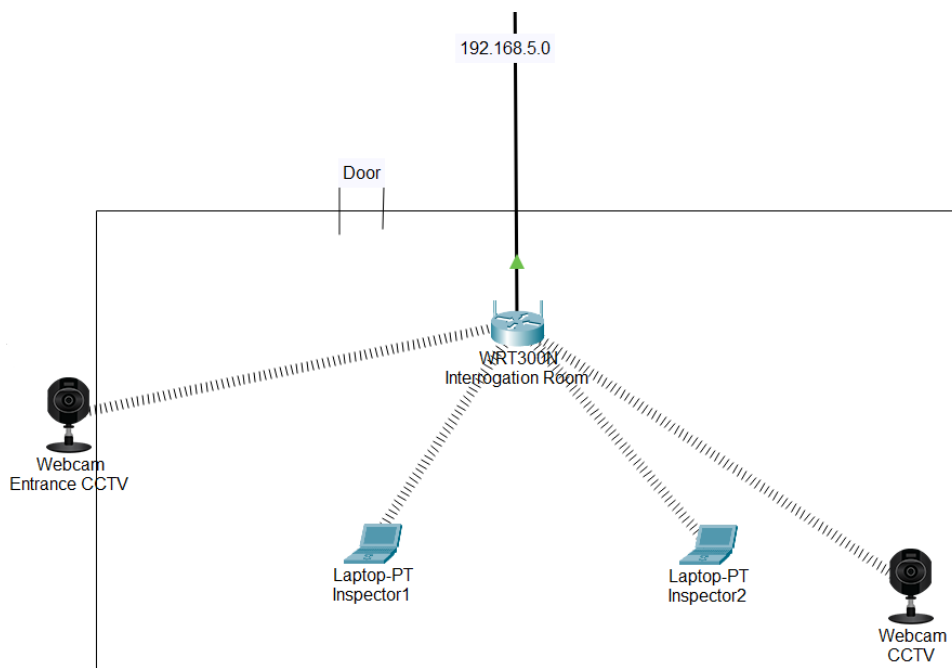
8. Save all made configurations:

“copy running-config startup-config”. Saves all new made configurations.

[5]

INTERROGATION ROOM

This room requires a DHCP or an access point to operate. This is where the suspects are put into interrogation. The room contains a wireless router as an access point and CCTV. When the officer comes inside, usually brings a laptop to work with. This department operate on 192.168.5.0 network, which is connected to the same router as officer's rooms above. Apart from the other devices, only the wireless router is configured with static IP address 192.168.5.2. The DHCP service is enable and configured with range of IP addresses start from 192.168.0.100 to 192.168.0.200.

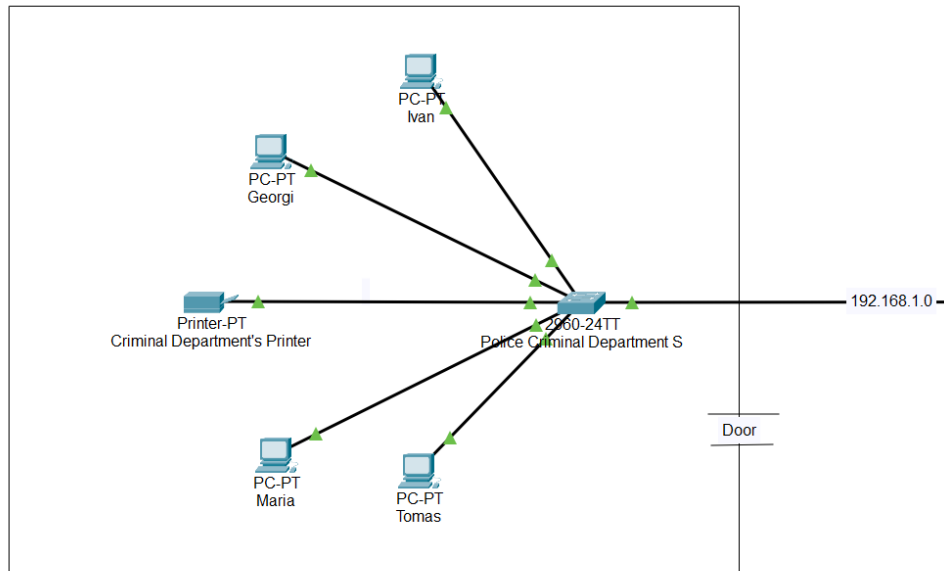


This is Cisco wireless router WRT300N

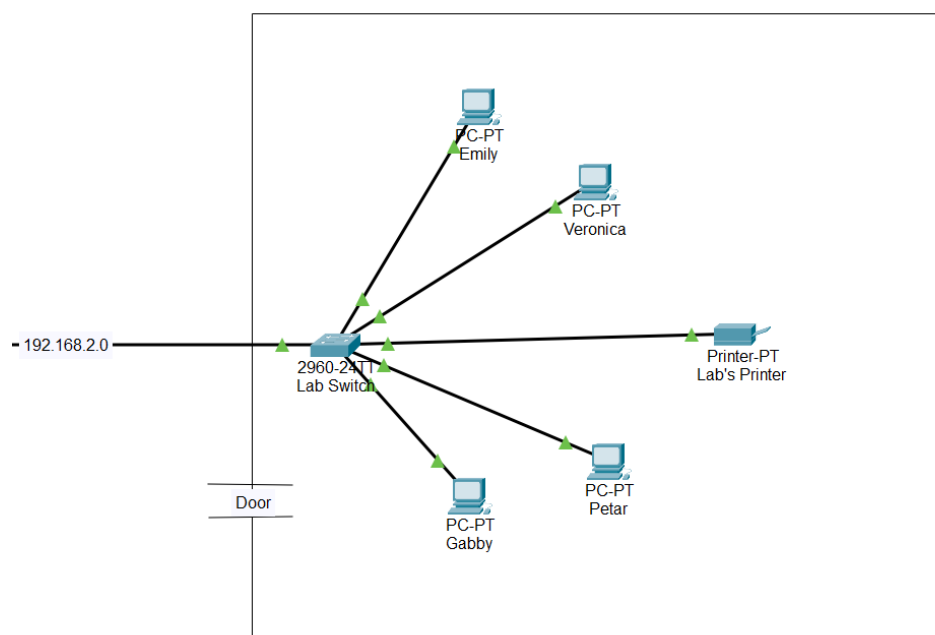


CRIMINAL DEPARTMENT & LAB DEPARTMENT

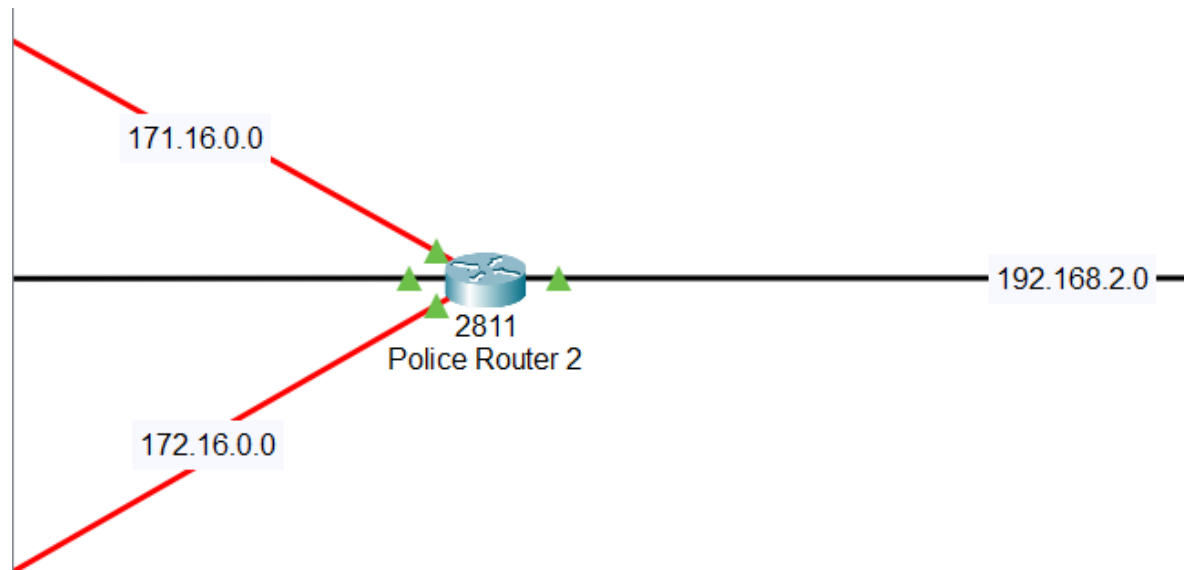
Criminal department is the room where advanced investigations and research take place. It is mostly used by the inspectors. This department is on network 192.168.1.0



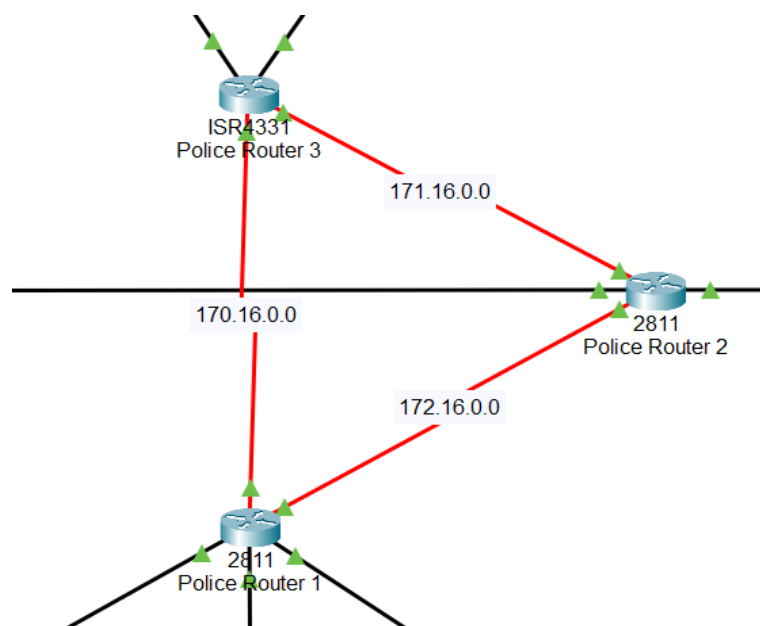
Lab department is where procedures like, fingerprint and DNA results take place. Department network is 192.168.2.0



Those departments are connected into a router, same as the one that connects the previous two departments.



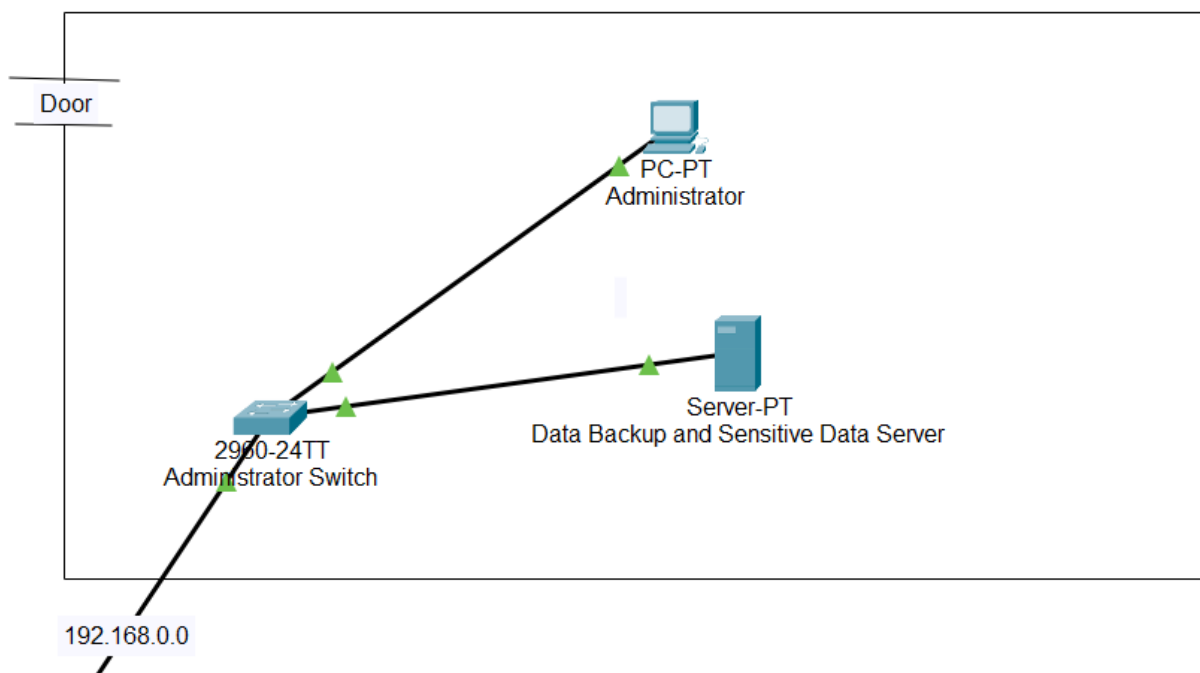
This router connects those two departments, and it is a part of redundancy strategy in form of triangle.



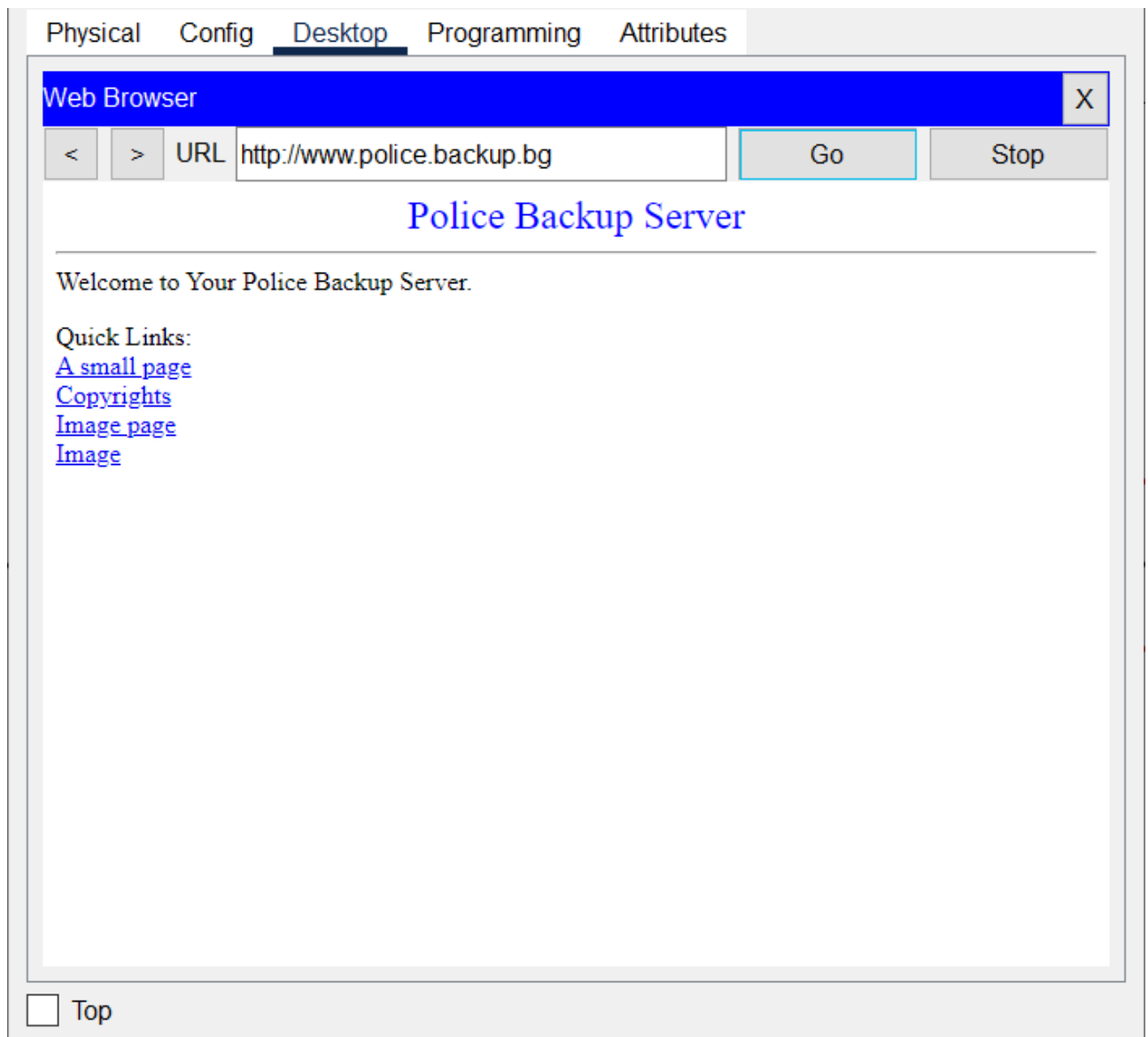
In case of faultier of one of the routers, the rest departments can still be able to communicate with each other.

ADMINISTRATOR ROOM

The administrator's room is for the person, responsible for maintaining the network and its users. He can do it from everywhere but, this project has the idea of building room only for this purpose. From the computer, the administrator can be able to manage all devices via SSH security channel, without the need for going to the particular device. Also, he or she will be responsible for managing the users with services like "Active Directory", depending on the system that will be used. By managing, mean set up users, passwords, policies, restricted access etc.



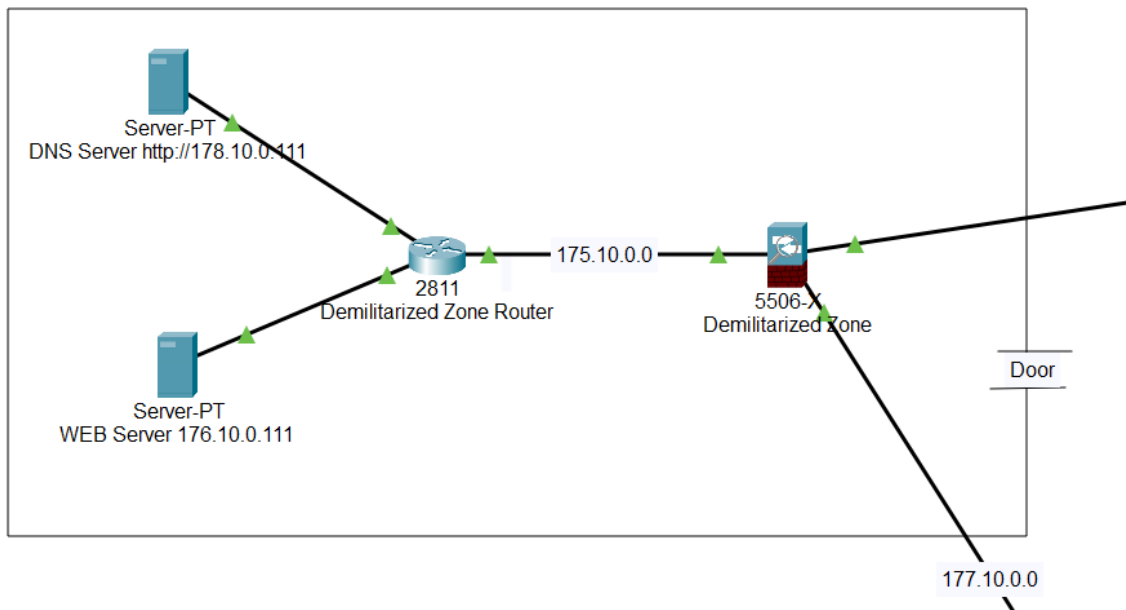
Also, the station's data backup and sensitive data server is placed in this room. This server can be supplied with hard drivers, where the data from the PCs will be copy and stored as a backup. Due to the fact that it is a police station, the data will be labeled as sensitive. Depending of the inspector rank, there will be authorized access to this server. The server runs its own application. Room network is 192.168.0.0.



This is the home page of the server's application, it has domain name in its server records, and in the DNS server. All PCs are configured with the DNS server's IP address as a DNS IP address.

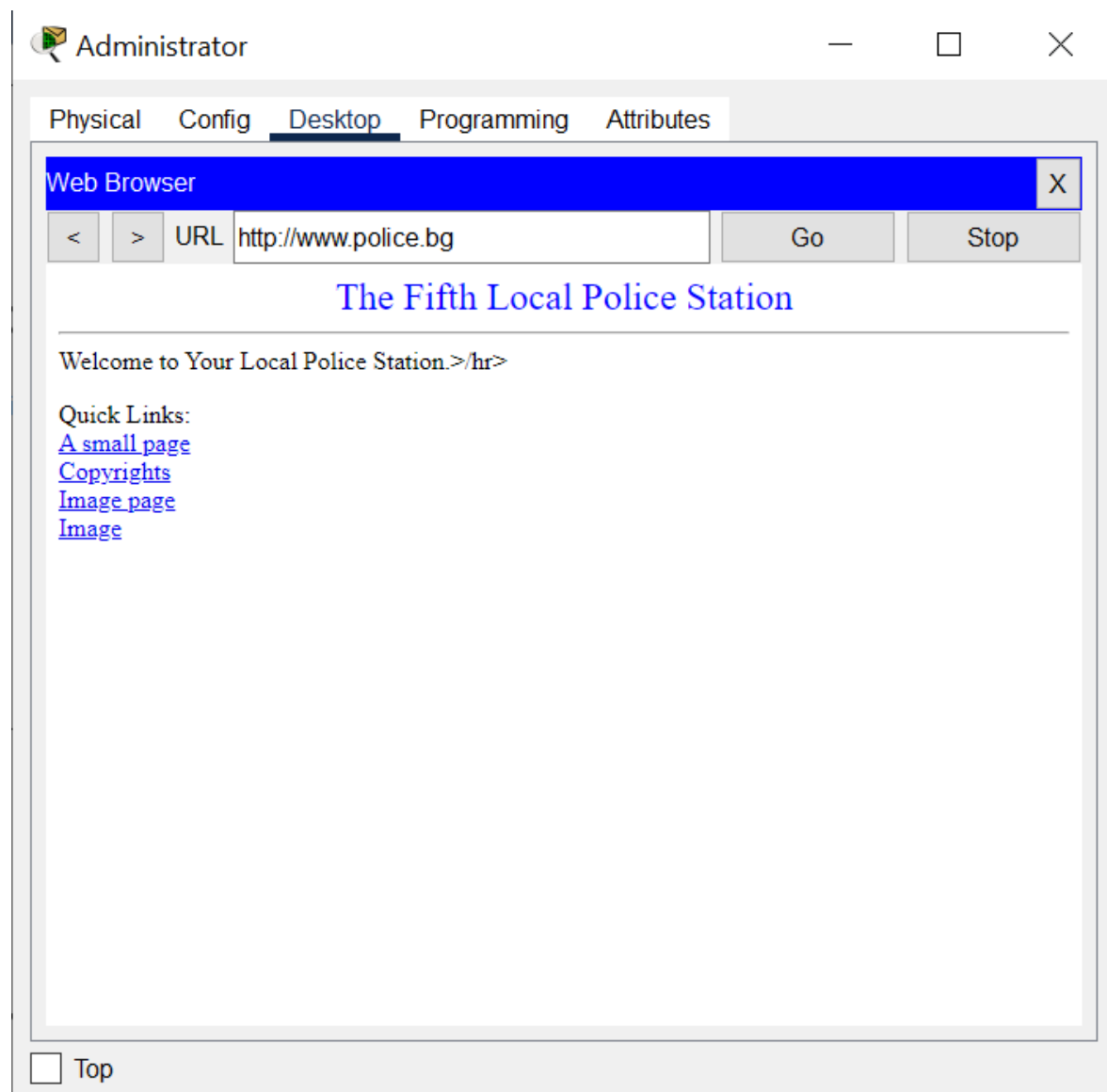
SERVER ROOM

In this room are two servers, the WEB server, and the DNS server. One important security component is there as well, the firewall that will play the role of DMZ (Demilitarized zone).



The firewall is configured with an IP address for each connected interface and a RIP routing protocol. The interface with network 177.10.0.0, is where the entire police local network is connected. This interface is configured with security level 100, which means that this interface is a 100% trusted, it is named “inside”. The server interface is named DMZ and it is with security level 70. The interface that faces the wild internet, is named “outside” with security level 0, 0% trusted. The firewall does not have a gateway of last resort defined to enable it to reach external networks. A static default route is configured on the firewall outside interface. If the firewall outside interface was configured as a DHCP client, it could obtain a default gateway IP address from the ISP. After this, a NAT protocol is used to translate the inside network addresses (177.10.0.0/16) to the global address of the outside firewall interface. In last, is the ICMP protocol, which will allow the traffic from inside, to be able to pass the outside interface as a return packed. Otherwise, the firewall will block it on the way back, even though the packet is allowed to go out through the outside interface. [19] [20]

The DNS server, keeps record of the Web server's location, domain name and IP address, the WEB server, runs an application, a web site where the users can see different information that has been posted, events, news etc. Both servers, can act as backup, if one of them fails.



VLAN (VIRTUAL LOCAL NETWORK)

ADDITIONAL PROJECT

Creating logically divided networks within a physical network architecture is possible thanks to a computer networking technique called a VLAN, or virtual local area network. Even if your devices aren't physically linked to the same network switch, you may still group them together based on numerous criteria, such as function, department, or security needs, using VLANs.

- 1. Logical Segmentation:** VLANs provide you the ability to construct virtual segments in place of only relying on physical connections. Even though they are physically apart, devices in the same VLAN can interact with one another as if they were on the same physical network.
- 2. Switch-Based Segmentation:** Network switches are commonly used to implement VLANs. On the same physical hardware, you may define and configure many VLANs using a managed switch.
- 3. VLAN tagging:** Network switches employ a method known as VLAN tagging to distinguish traffic from various VLANs. Each data packet's header receives a tag when it leaves a device, identifying the VLAN to which it belongs. Switches use this tag to correctly route traffic to the appropriate VLANs.
- 4. Network isolation and security** are two of VLANs' main advantages. By default, devices in different VLANs are segregated from one another. By prohibiting unauthorised communication between devices that shouldn't talk to one another, this can increase security.
- 5. ARP queries and other broadcast traffic** are delivered to any device connected to a traditional LAN, which shares a broadcast domain. By limiting the scope of broadcast communication to just devices in the same VLAN, VLANs help cut down on unneeded network traffic.
- 6. Prioritising traffic:** Some switches include Quality of Service (QoS) capabilities that let you give particular VLANs and types of traffic top priority. In a Voice VLAN, for instance, you may give VoIP traffic priority over standard data traffic.

7. Flexibility in network administration is made possible through VLANs. Without having to physically rewire connections, you may quickly reconfigure and modify your network to meet changing requirements.

Common uses of VLANs include:

Departmental Segmentation: Different departments in an organization can be placed in separate VLANs, improving network organization and security.

Guest Networks: VLANs can be used to isolate guest devices from the main network while still providing internet access.

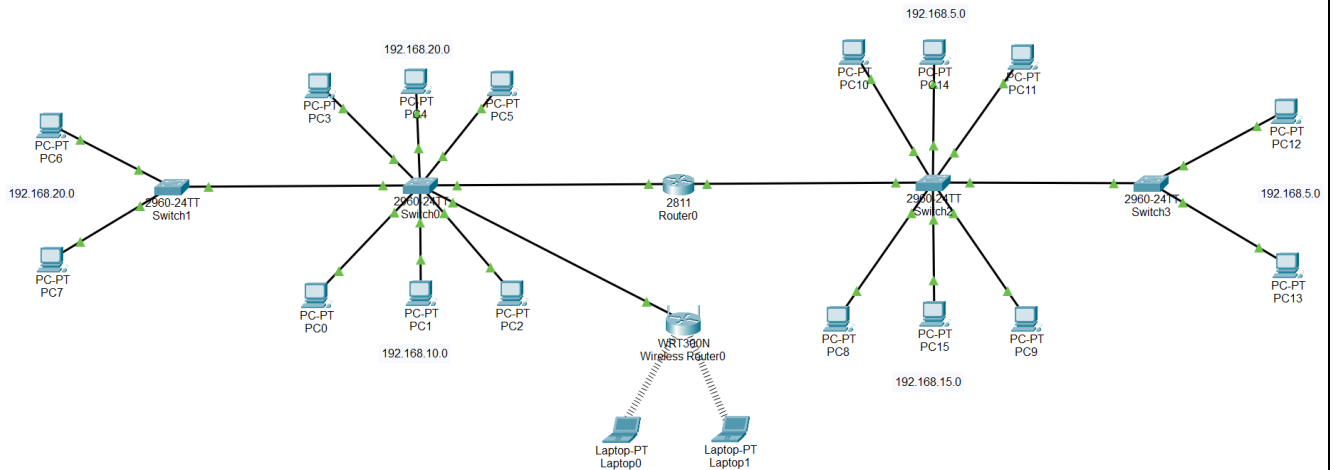
IoT Devices: Devices like smart TVs, smart thermostats, and security cameras can be placed in their own VLAN to enhance security.

VoIP Networks: Voice over IP (VoIP) phones can be placed in a dedicated VLAN to ensure call quality and prioritize voice traffic.

VLANs are an important tool for network administrators to optimize network performance, security, and organization in complex network environments.

[17]

Based on the research and the chat with the inspector, this particular station does not need VLAN at this point, due to the fact that it is not big organisation. However, this project is ready with a plan for VLAN example topology, in case of need for the future or another bigger police station.



This VLAN example, has a router connecting four different networks, that are on a VLAN management. The VLAN on the right side, network 192.168.15.0 operates on VLAN 15 on the switch, which is configured to be in a range of 20th to 24th port in the switch. This network is isolated from 192.168.5.0 network and yet they both are connected to the same router. The same goes for the devices on the left side. Every device is able to ping its neighbours on the same network, plus its default gateway but, cannot ping the devices from the other networks. The router has its fast ethernet interfaces divided into two channels, this is called router-on-a-stick. Those interfaces are configured with 802.1Q standard protocol. VLAN is also suitable for multi switching, which will decrease the cost of devices. There is Wi-Fi option as well, in case of need, connected to VLAN network 10, fully able to ping the main router.

A router-on-a-stick is an inter-VLAN routing technique in which the router and switch are linked by a single physical interface, thus the name. This technique is used in the majority of contemporary inter-VLAN routing implementations. Router-on-stick does not require numerous physical interfaces on both the router and the switch, in contrast to the conventional inter-VLAN routing technique. Instead, it is possible to configure the router's operating system to operate the router interface as a trunk link, which is then connected to a switch port that is set up in trunk mode. This suggests that switching between different VLANs may be accomplished using a router and switch with a single

physical interface. Multi-vendor VLAN support is provided via the IEEE 802.1Q (Dot1q) protocol, which establishes a system of VLAN tagging for Ethernet packets. Several proprietary protocols, including Cisco Inter-Switch Link (ISL) and 3Com's Virtual LAN Trunk (VLT), were in use before the 802.1Q standard was introduced. [18] [20]

Those are the basic commands and steps for creating a VLAN:

Enter privileged EXEC mode:

“enable”.

Configure VLANs:

“configure terminal”.

Create VLANs:

“vlan 10”.

“name FirstDepartment”.

“vlan 20”.

“name SecondDepartment”

Assign VLANs to switch ports:

“interface range FastEthernet0/1-11”. Enters the configuration on the block of ports 1 till 11.

“switchport mode access”. Goes to access mod.

“switchport access vlan 10”. Makes all those ports on VLAN 10.

“interface range FastEthernet0/12-24”. Enters the configuration on the block of ports 12 till 24.

“switchport mode access”. Goes to access mod.

“switchport access vlan 20”. Makes all those ports on VLAN 20.

“interface GigabitEthernet0/1”. Goes to gigabit interface 0/1

“switchport mode trunk”. Makes this interface to act as a trunk. This is the interface, where the switch is connected to the router.

[5]

SUMMARY

Local Area Network (LAN) Project Summary

Project Overview:

The Local Area Network (LAN) Project aims to enhance the connectivity, performance, and security within the station. The project involves network components, implementing advanced security measures, and optimizing network resources to ensure seamless communication and data transfer among devices within the local network.

Objectives:

Scalability: Design the LAN architecture with scalability in mind, accommodating future growth in the number of devices and users without compromising performance.

Security Strengthening: Implement robust security measures, including firewall configurations, to safeguard sensitive data and protect against unauthorized access.

Network Monitoring and Management: Integrate network monitoring tools to enable real-time monitoring of network health, traffic patterns, and device status. Implement centralized network management for easier administration and troubleshooting.

Redundancy and High Availability: Set up redundant network paths and failover mechanisms to ensure uninterrupted network connectivity even in case of hardware failures.

Quality of Service (QoS) Implementation: Prioritize network traffic based on application and user requirements, ensuring critical applications receive the necessary bandwidth for optimal performance.

The Local Area Network Project is poised to significantly enhance the organization's network capabilities, providing improved performance, security, and scalability. Through careful planning, diligent implementation, and ongoing maintenance, the LAN will support the organization's current and future technological needs.

REFERENCES

[1] Enterprise Network and security management

second edition 2022

Bongsik Shin

[2] Networking All-in-one

7th edition 2018

Doug Lowe

[3] IEEE Journal on Selected Areas in Communications (Volume: 9, Issue: 7, September 1991)

[4] Wide Area Network Design: Concepts and Tools for Optimization. Robert S. Cahn

[5] Cisco Network Academy

[6] Internet Connectivity, Community Participation, and Place Attachment: A Longitudinal Study

Gustavo S. Mesch and Ilan Talmud

Volume 53, Issue 8

[7] Security analysis in role-based access control

Ninghui Li, Mahesh V. Tripunitara

ACM Transactions on Information and System Security Volume 9, Issue 4

01 November 2006

[8] IEEE Potentials (Volume: 20, Issue: 1, Feb-March 2001)

[9] IEEE Internet Computing (Volume: 3, Issue: 4, July-Aug. 1999)

[10] IEEE Communications Magazine (Volume: 45, Issue: 10, October 2007)

[11] Interconnecting Cisco Network Devices, Part 1 (ICND1) Foundation Learning Guide 2013

[12] Network Devices a Complete Guide - 2021 Edition

Publisher: The Art of Service - Network Devices Publishing (17 Dec. 2020)

[13] Network Cabling Illuminated

Robert Shimonski, Richard T. Steiner, Sean M. Sheedy

2006

[14] Guide to Network Cabling Fundamentals

Beth Verity

2003

[15] Packet Tracer Network Simulator

Jesin A.

2014

[16] Configuration management for IP network devices

IEEE

2004

[17] Improving LAN Performance Based on IEEE802.1Q

VLAN Switching Techniques

Dhurgham Abdulridha Jawad AL-Khaffaf

[18] Inter-VLAN Routing Configuration Tutorial

AMAKIRI WELEKWE TECHNOLOGY ADVISOR | CYBERSECURITY EVANGELIST

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[19] CCNA Security

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[20]

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